

EE Outline

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Title:

Investigating the Effect of Asynchronous I/O on Web Server Performance: aiohttp versus Flask under Increasing Concurrent Loads

Research Question:

To what extent does asynchronous I/O using aiohttp improve server response time compared to synchronous Flask under increasing concurrent loads?

1. Abstract (150–200 words)

Overview of synchronous and asynchronous I/O paradigms in Python web servers. Description of experiment, comparing Flask and aiohttp, with metrics including average response time, 95th percentile latency, throughput, and CPU usage. Summary of methodology, anticipated results, and implications for scalable web applications.

2. Intro (250–300 words)

Context: Synchronous frameworks like Flask process requests sequentially, leading to bottlenecks under high concurrency. Asynchronous alternatives like aiohttp use non-blocking I/O via asyncio event loops.

Hypothesis: High-concurrency technologies are prevalent in modern web services. Aiohttp is expected to demonstrate better performance, reducing response times at higher loads and similar results at lower loads.

3. Theory (300–350 words)

Concepts:

- Synchronous I/O: Thread-blocking model limited by Python GIL.
- Asynchronous I/O: Coroutine-based concurrency.

Performance Metrics:

- Average response time ($t_{avg} = \frac{\sum t_i}{n}$).
- 95th percentile latency.
- Throughput (requests/second).
- CPU utilization (%).

Literature: Official documentation (Flask, aiohttp), benchmarks from TechEmpower Framework Benchmarks.

4. Methodology (500–600 words)

Experiment Design: Quantitative controlled trial.
Independent variable: concurrent requests (10, 50, 100, 500, 1000). **Dependent variables:** specified metrics.
Apparatus:

Component	Synchronous (Flask)	Asynchronous (aiohttp)
Framework Version	Flask	aiohttp
Server Mode	Development server	asyncio event loop
Load Generator	Apache Bench (ab)	Apache Bench (ab)
Endpoint	GET /api/data (JSON)	GET /api/data (JSON)

Environment: Arch Linux, Python 3.12, localhost.

Procedure:

- 1. Implement minimal reproducible servers.
- 2. Execute five trials per load level.
- 3. Record metrics using `time.perf_counter()` and `psutil`.

Controls: Identical request payloads

5. Results and Analysis (500–600 words)

Data Presentation:

Concurrent Load (req/s)	Flask Avg. Response Time (ms)	aiohttp Avg. Response Time (ms)	Throughput (req/s)	Improvement (%)	95th Latency (ms)	CPU Usage (%)
10	-	-	-	-	-	-
1000	-	-	-	-	-	-
10000						
More if needed						

Visualizations: Line graphs for response time vs. load; error bars (± 1 SD).
Generated via python library Seaborn
Analysis: Quantitative comparison, correlation coefficients, significance testing.

6. Discussion (250–300 words)

Interpretation: Alignment with hypothesis,
explanation via event loop efficiency.

Implications: Relevance to production web systems.

Limitations: Single-machine testing (localhost): no network latency ; The Research investigates difference not exactly in async or sync but it specific technologies as Flask but aiohttp

For Further Research: Hybrid models, distributed environments.

7. Conclusion (150–200 words)

Restatement of research and experiment results relative to the hypothesis.

Contribution to understanding asynchronous paradigms in computer science.

Bibliography (Harvard style): 10–15 sources including primary documentation.

Appendices: Source code, raw datasets, supplementary figures.

Secondary sources:

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